

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – COMPARATIVE ANALYSIS (5 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization (BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO4 Statement:	Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).		
Student Name:	G Kushwanth	Reg. No.:	192424150
Section / Batch:	AI&DS (2024 Batch)	Date:	

Code of Conduct

*I **G Kushwanth 192424150** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Scatterplots & Hexbin Plots	Comparative analysis of scatterplots and hexbin plots for large-scale clustered data visualization, density estimation, and overplotting handling	1	10
Dimension Reduction Techniques	PCA, t-SNE, and UMAP comparison for high-dimensional data reduction, clustering, and structure preservation	2	10
Correlograms & Network Graphs	Correlation visualization, threshold-based network construction, and multivariate relationship analysis	3	10

Time Series Analysis & Decomposition	Moving Average, Detrending, and STL decomposition for trend extraction, seasonality analysis, and forecasting suitability	4	10
Geospatial Data Visualization	Choropleth maps, cartograms, and geospatial heatmaps for spatial pattern analysis and hotspot detection	5	10
GRAND TOTAL:		50 Marks	

Annexure A – Comparative Analysis

1. A retail company maintains transaction records containing customer spending amounts. Construct a Histogram, Density Plot, and ECDF (Empirical Cumulative Distribution Function) and perform a comparative analysis.

Analysis Parameters:

- A. Distribution shape identification
- B. Detection of skewness and outliers
- C. Representation of cumulative information
- D. Sensitivity to sample size
- E. Ease of interpretation **Answer:**

The Histogram bins spending values into intervals and shows raw frequency counts, making it simple but sensitive to bin width and sample size. The Density Plot smooths the same data using kernel density estimation (KDE), producing a continuous curve that reveals the underlying distribution shape without binning artifacts. The ECDF plots the cumulative proportion of observations at or below each value, giving an exact, assumption-free view of the distribution and enabling direct probability and percentile reads.

Parameter-wise Evaluation:

- A. Distribution shape: Density Plot best reveals modality and skewness smoothly; Histogram shows it coarsely depending on bin count; ECDF shows shape only indirectly through curve steepness.
- B. Skewness/outliers: Histogram and Density Plot visually flag skew and tail outliers; ECDF flags outliers as isolated points far from the main curve trajectory.
- C. Cumulative information: only the ECDF directly encodes cumulative proportions; Histogram and Density Plot require visual integration.
- D. Sensitivity to sample size: Histogram is highly bin/size sensitive; Density Plot depends on bandwidth choice; ECDF is the most robust and sample-size stable.
- E. Interpretability: Histogram is most intuitive for general audiences; Density Plot is intuitive for shape comparison; ECDF requires more statistical literacy but is most precise.

Example/Application: For customer spending data, a Density Plot quickly shows whether spending is right-skewed with high-value outliers, while the ECDF can answer precise questions such as "what fraction of customers spend below Rs. 5000," which the Histogram cannot answer directly.

2. A healthcare dataset contains patient attributes such as age, blood pressure, cholesterol, and BMI. Generate a Pair Plot, Parallel Coordinates Plot, and Scatterplot Matrix and compare their effectiveness in multivariate analysis.

Analysis Parameters:

A. Relationship identification among variables

B. Detection of clusters and anomalies

C. Scalability with increasing dimensions

D. Visual complexity and readability E. Suitability for exploratory data analysis **Answer:**

A Pair Plot arranges scatterplots for every variable pair in a grid with univariate distributions on the diagonal, making it effective for small-to-moderate variable counts. A Parallel Coordinates Plot represents each variable as a parallel vertical axis and each observation as a connected line across axes, scaling better to more dimensions. A Scatterplot Matrix is structurally similar to a Pair Plot but typically omits diagonal distribution plots, focusing purely on pairwise correlation patterns.

Parameter-wise Evaluation:

A. Relationship identification: Pair Plot and Scatterplot Matrix directly show pairwise linear/non-linear relationships; Parallel Coordinates shows relationships as line-crossing patterns, which is less precise for two-variable correlation.

B. Clusters/anomalies: Pair Plot is strongest for detecting clusters via point grouping; Parallel Coordinates reveals clusters as bundles of near-parallel lines and anomalies as crossing outlier lines.

C. Scalability with dimensions: Parallel Coordinates scales best to many variables (age, BP, cholesterol, BMI, etc.); Pair Plot and Scatterplot Matrix grow quadratically and become cluttered beyond 6-8 variables.

D. Visual complexity: Pair Plot/Scatterplot Matrix panels stay individually simple but the overall grid gets dense; Parallel Coordinates suffers from line-overplotting with large datasets.

E. Exploratory suitability: Pair Plot is best for initial exploratory correlation screening; Parallel Coordinates is best for comparing multiple patient profiles simultaneously.

Example/Application: In the patient dataset, a Pair Plot quickly reveals that BMI and blood pressure are positively correlated, while a Parallel Coordinates Plot can visually flag a patient whose combination of high cholesterol and low BMI deviates from the typical pattern.

3. A manufacturing dataset contains measurements from multiple sensors. Create a Heatmap, Correlogram, and Clustered Heatmap to analyze relationships among variables and perform a comparative analysis.

Analysis Parameters:

A. Correlation representation

B. Pattern and anomaly detection

C. Grouping of related variables

D. Scalability to large feature sets

E. Interpretability of results

Answer:

A Heatmap uses color intensity in a matrix to represent correlation coefficients (or raw sensor values) between variable pairs, making strong relationships instantly visible. A Correlogram builds on this by combining correlation coefficients with visual encodings such as ellipses, circles, or color gradients, often ordered to group similar variables. A Clustered Heatmap adds hierarchical clustering

(dendrograms) to reorder rows and columns, automatically grouping sensors with similar behavior together.

Parameter-wise Evaluation:

- A. Correlation representation: all three encode correlation via color; the Clustered Heatmap additionally reorders variables for clearer grouping.
- B. Pattern/anomaly detection: Clustered Heatmap is most effective, since anomalous sensors appear as isolated rows/columns outside main clusters.
- C. Grouping related variables: only the Clustered Heatmap automatically groups correlated sensors via dendrograms; plain Heatmap and Correlogram require manual inspection.
- D. Scalability: basic Heatmap scales well to many sensors but becomes visually dense; Correlogram is best for a moderate number of variables where symbol shapes matter; Clustered Heatmap remains interpretable at scale due to grouping.
- E. Interpretability: Correlogram is most intuitive due to shape/color combination; Clustered Heatmap requires familiarity with dendrograms but gives the deepest structural insight.

Example/Application: In the manufacturing dataset, a Clustered Heatmap can reveal that vibration and temperature sensors cluster together as one group while pressure sensors form a separate group, helping engineers isolate correlated fault sources.

4. A financial time-series dataset contains daily stock prices recorded over several years. Apply Line Plot, Moving Average Plot, and STL Decomposition and perform a comparative analysis.

Analysis Parameters:

- A. Trend identification**
- B. Seasonality detection**
- C. Noise reduction capability**
- D. Responsiveness to changes**
- E. Suitability for forecasting Answer:**

A Line Plot shows the raw stock price series over time, preserving all detail including noise, making short-term fluctuations and long-term trends both visible but sometimes hard to separate. A Moving Average Plot smooths the series by averaging over a rolling window, reducing noise to highlight the underlying trend at the cost of lag and loss of short-term detail. STL Decomposition (Seasonal-Trend decomposition using Loess) formally separates the series into trend, seasonal, and residual components, giving the clearest structural breakdown.

Parameter-wise Evaluation:

- A. Trend identification: STL Decomposition isolates the trend component explicitly; Moving Average approximates it; the raw Line Plot only shows trend implicitly beneath noise.
- B. Seasonality detection: only STL Decomposition explicitly extracts a seasonal component; Line Plot and Moving Average can only hint at seasonality through repeated visual patterns.
- C. Noise reduction: Moving Average reduces noise well but introduces lag; STL separates noise into a residual component without distorting the trend; the Line Plot has no noise reduction.
- D. Responsiveness to changes: Line Plot reacts instantly to price changes; Moving Average is deliberately less responsive (lag increases with window size); STL responds accurately since trend and season are modeled separately.
- E. Forecasting suitability: STL Decomposition is most suitable, as its clean trend and seasonal components feed directly into forecasting models; Moving Average offers basic short-term forecasting; the raw Line Plot is not directly forecast-ready.

Example/Application: For a multi-year stock price series, STL Decomposition can separate a genuine upward trend from a recurring quarterly earnings-driven seasonal pattern, something a plain Line Plot or Moving Average would blur together.

5. A social network dataset contains interactions among users. Construct a Network Graph, ForceDirected Graph, and Adjacency Matrix Visualization and compare their effectiveness.

Analysis Parameters:

A. Relationship representation

B. Community detection capability

C. Scalability with network size

D. Visual clutter management E. Identification of influential nodes **Answer:**

A Network Graph represents users as nodes and interactions as edges using a fixed or algorithmic layout, showing overall connectivity structure. A Force-Directed Graph is a specific, physics-based layout of a network graph where connected nodes attract and unconnected nodes repel, naturally revealing clusters and central nodes through spatial positioning. An Adjacency Matrix Visualization instead encodes the same connections as a grid, with cell shading indicating the presence or strength of interaction between each pair of users, avoiding node-link overlap entirely.

Parameter-wise Evaluation:

A. Relationship representation: Network Graph and Force-Directed Graph represent relationships intuitively as visible links; Adjacency Matrix represents them more abstractly as matrix cells.

B. Community detection: Force-Directed Graph is strongest, since spatial clustering emerges naturally from the physics simulation; Adjacency Matrix requires row/column reordering to reveal communities. C. Scalability with network size: Adjacency Matrix scales best for very large, dense networks since it avoids edge-crossing clutter; Network and Force-Directed Graphs become visually tangled ("hairballs") at scale.

D. Visual clutter management: Adjacency Matrix manages clutter best for dense networks; Force-Directed Graph manages it reasonably well for sparse-to-medium networks; a basic Network Graph clutters quickly.

E. Identification of influential nodes: Force-Directed Graph is most effective, as highly connected (influential) users visually appear larger or more central; Adjacency Matrix shows influence as denser rows/columns, which is less immediately visible.

Example/Application: For a social network dataset, a Force-Directed Graph can visually highlight an influencer user as a densely connected hub, while an Adjacency Matrix would be preferable if the platform has millions of users, where node-link diagrams would become unreadable.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks

Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant		
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison		
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided		
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation		
Clarity of Presentation	15%	Well organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____